Clinical Goal

To design a spinal cage that supports early mechanical load, promotes bone formation, adapts over time, enables radiological tracking, and maintains patient safety.

Final Cage Design Structure

✅ Ti Outer Shell: Provides strong, permanent mechanical support. Placed on the apophseal ring (cortical bone), ideal for screw fixation. 3D printed and surface-treated for bioactivity.

✅ Slot-in Mg Alloy Core: Biodegradable insert aligned with the cancellous bone. Allows adaptive modulus through gradual degradation. Chosen intraoperatively as fast, medium, or slow degrading core.

✅ Chitosan Coating: Applied to Mg core to delay degradation and match bone healing rate. Prevents premature structural loss.

✅ rhBMP-2 Bone Graft Chamber: Central slot filled at surgery with rhBMP-2 to promote active osteoinduction. Sterilisation must be managed carefully—BMP added in sterile field as gamma sterilisation can denature it.

Optional enhancements include permanent surface charge to attract growth factors, and piezoelectric PVDF layers to stimulate osteoblasts via micromovement.

Component Summary Table

Component	Material	Purpose
Outer layer	3D printed Titanium	Strong, permanent support; interfaces with apophseal ring
Inner core	Slot-in Mg alloy, coated	Resorbable; allows adaptive modulus and load transfer
Bone graft chamber	Filled with rhBMP-2	Stimulates new bone growth via osteoinduction
Surface features	Concave + nanoscale	Enhance osteoinduction and osteointegration
Optional coatings	Silver/copper, PVDF	Antimicrobial protection and electrical stimulation
Manufacturing	SLM + post-processing	Precision print, etched osteoactive surfaces, sterilised by gamma

To treat this, spinal fusion is performed: the degenerated disc is removed and replaced with an **interbody fusion cage**, which encourages bone growth between adjacent vertebrae. However, current designs often:

- **Shield the bone from stress** (causing bone weakening),
- Are **too stiff**, preventing bone from taking load (violating Wolff's Law),
- And/or **block imaging**, making fusion progress difficult to assess.

What is Wolff's Law? - Bone becomes stronger when it's used, and weaker when it's not.

- When your bones are **regularly loaded with weight or stress** (e.g., walking, running, lifting), they respond by **getting stronger**.
- If bones don't experience stress (e.g., you're lying in bed for weeks, or using a very stiff implant), they become **weaker and less dense**.

Mechanical Performance & Safety

- The cage must initially support **spinal load** during typical movements like bending or lifting.
- The **Apophseal ring**, composed of strong cortical bone, should interface with the **load-bearing parts** of the cage.
- **Cancellous bone** in the vertebral body center is softer and should be paired with **resorbable, less stiff materials**.

Final Cage Design

Outer Layer – Titanium

• Material: 3D printed Titanium (Ti)

• Location: Apophseal ring (cortical bone)

• Purpose: Provides strength, allows screw fixation, and supports early loading.

• Surface treatment: Acid/alkaline etching to introduce nanoscale features for osteoblast adhesion.

Inner Layer – Magnesium Insert

• Material: Resorbable Magnesium alloy (Mg)

• Purpose: Enables load transfer over time (adaptive modulus), bioactive, radiolucent

• Issue: Degrades too quickly

• Solution: Coated with chitosan to control degradation

• Modular inserts: Surgeon chooses fast, medium, or slow degrading core

Bone Growth Features

• Porosity and lattice structures allow vascularisation and reduce stiffness

• Concave pore geometry mimics bone remodeling sites (osteoclast pits)

• Submicron/nanoscale features increase protein adsorption and cell attachment

• Central chamber allows addition of rhBMP-2 for osteoinduction

• Permanent surface charge helps bind endogenous growth factors

• Optional piezoelectric coating (e.g. PVDF) enhances healing through microcurrent stimulation

Bone Graft Chamber (in centre of cage):

- Can be filled during surgery with **rhBMP-2**, which:
    1. **Stimulates bone growth**.
    2. **Provides scaffold** for osteoblasts.
    3. **Promotes fusion** between vertebra

Structural Design for Safety:

- **Porosity and Lattice Architecture**:
    - Reduce stress shielding.
    - Improve nutrient flow and integration.
- **Antimicrobial Coatings**:
    - **Silver**: Prevents infection, safe for bone regrowth.
    - **Copper**: May enhance angiogenesis, but used cautiously.

Driving Bone Formation

📏 **Porosity and Shape**

- **Porous lattice structures** improve nutrient flow, reduce stiffness, and allow bone ingrowth.
- **Concave pores** mimic **osteoclast resorption pits**, promoting osteoblast activity — a **biomimetic design** proven to induce bone even in non-bony areas.

🦠 **Surface Features**

- **Submicron/nanoscale roughness** increases surface area, protein adsorption, and cell attachment.
- Achievable via **acid etching on titanium** post-printing.

🧬 **Biological Additions**

- Central **bone graft chamber** to load **rhBMP-2** at surgery, encouraging bone growth (osteoinductive).
- Optional addition of **piezoelectric materials** (e.g. **PVDF**) that create microcurrents during movement, enhancing osteogenesis.

Radiolucency

- **Imaging is essential** to assess bone fusion.
- **Metals (Ti)** cause artefacts; **PEEK** and **Mg** are more radiolucent.
- Best approach: an **adaptive radiolucent design** where inner resorbable parts (e.g. Mg) **degrade over time**, making fusion more visible.

• Titanium is radiopaque but provides structural support

• Magnesium is radiolucent and degrades over time

• Adaptive radiolucency: structure becomes more visible as Mg degrades, allowing imaging of bone healing

Antimicrobial Coatings

- **Silver**: Safe in small doses, prevents biofilm, used in trauma/dental implants.
- **Copper**: May stimulate angiogenesis by mimicking hypoxia but less studied.
- Optional use depending on infection risk.

Manufacturing Summary

• Titanium cage made using Selective Laser Melting (SLM)

• Post-processing includes heat treatment, acid etching, and gamma sterilisation

• Magnesium insert machined separately and coated with chitosan

• Modular slot-in design allows intraoperative customisation

Material Comparison

Material	Pros	Cons
Titanium	Strong, good osseointegration, suitable for screws	Radiopaque, stress shielding if solid
Magnesium	Degradable, bioactive, radiolucent	Fast degradation, must be coated
PLA	Biodegradable, low cost	Produces acidic by-products, poor cell response
PEEK	Radiolucent, lightweight	Hydrophobic, poor osseointegration
CFR-PEEK	Strong, light	Not 3D printable, fibre migration risk

Q1. Explain how the spinal cage design ensures safe load bearing while bone heals.

Model Answer:

The spinal cage achieves safe load-bearing through a hybrid design combining strength and adaptability:

- **Outer Ti Shell:** Made via SLM, this 3D printed titanium cage interfaces with the **apophseal ring**, the strongest cortical bone, ensuring high mechanical stability. Titanium supports screw fixation and resists long-term deformation.
- **Inner Mg Alloy Core:** A **resorbable magnesium insert** is slotted inside, aligned with the softer cancellous bone. It provides early support but degrades over time, transferring load to the new bone – fulfilling the **adaptive modulus** goal.

- **Coating with Chitosan:** Prevents Mg from degrading too quickly, ensuring the cage doesn't lose structural integrity before bone has formed.
- **Modular Design:** The Mg insert is available in fast/medium/slow degrading versions to allow **personalisation at point of use**, matching patient healing rate.

This structure mimics natural load transfer, avoids stress shielding (Wolff's Law), and promotes gradual, safe load-bearing.

Q2. How does the cage design promote bone formation?

Model Answer:

The spinal cage enhances bone growth using several biomimetic and bioactive strategies:

- **Porous Structure:** Allows vascularisation and movement of cells, nutrients, and waste – critical for healing.
- **Concave Pores:** These biomimetic shapes mimic osteoclast resorption pits. Osteoblasts are drawn to them, promoting **osteointegration** even in unloaded areas.
- **Submicron/Nanoscale Features:** Created via acid etching on Ti, these increase protein adsorption and **cell attachment**, enhancing **osteoconduction**.
- **Surface Coatings:** Optional nanoscale Ti or HA coatings improve bone-on growth; permanent surface charge can attract growth factors.
- **Bone Graft Chamber with rhBMP-2:** Positioned centrally, it triggers **osteogenesis** (bone formation from undifferentiated cells) – especially useful in areas with limited natural bone.

Together, these elements promote **osteoconduction**, **osteointegration**, and **osteoinduction**.

Why are Ti and Mg chosen over materials like PEEK or PLA?

Model Answer:

The Ti-Mg combination offers optimal strength, bioactivity, and radiolucency:

- **Titanium (Ti):** Strong, stiff, easy to 3D print and post-process (acid/alkaline etching), creates nano-rough surfaces that osteoblasts adhere to. Best for outer shell and screw fixation.
- **Magnesium (Mg):** Bioactive, resorbable, radiolucent. Degrades over time to allow adaptive modulus. Coated with chitosan to slow degradation and maintain strength.

Compared to alternatives:

- **PLA:** More radiodense (can hide bone healing), acidic degradation, poor osteoblast response.
- **PEEK:** Radiolucent but hydrophobic and hard to form porous structures. Poor osteointegration.
- **CFR-PEEK:** Stronger, but can't be 3D printed easily and has risk of fibre migration if not fully embedded.

Thus, Ti+Mg offers the best combination for mechanical stability, biological response, and imaging compatibility.

Q4. How does the cage allow radiographic monitoring of bone formation?

Model Answer:

The design achieves **adaptive radiolucency** for real-time monitoring:

- **Titanium Shell:** Radiopaque, provides initial stability and visibility on X-ray.
- **Magnesium Core:** Radiolucent and resorbable. As it degrades, the implant becomes more transparent to X-rays, revealing new bone formation in the core.
- This staged transparency allows **progressive imaging** of healing without removing the implant.
- **PEEK and PLA** are more radiolucent but lack bioactivity; the Ti-Mg design balances both visibility and biological performance.

This ensures clinicians can assess healing during the critical 6–12 week post-op period without interference from artefacts.

Q5. Describe how manufacturing techniques are used to produce this cage.

Model Answer:

The cage uses **hybrid manufacturing** to optimise structure, strength, and bioactivity:

- **Outer Titanium Shell:**
  - Made by **Selective Laser Melting (SLM)** or **DMLS**.
  - CAD model is printed layer-by-layer using Ti-6Al-4V powder.
  - Post-processing includes:
    - Heat treatment
    - Surface roughening (acid etching, anodisation)
    - Gamma sterilisation
- **Inner Magnesium Insert:**
  - Machined from solid Mg bar (safe and mature method).
  - Coated with **chitosan** via dip-coating to slow degradation.
  - Available in modular variants (fast/med/slow degradation rates).
- **Final Assembly:**
  - Slot-in design allows the Mg insert to be inserted intraoperatively.
  - Surgeons select insert based on patient-specific healing needs.

The combination allows precision, strength, and biological customisation in a clinically ready format.

How does the spinal cage design ensure patient safety while promoting bone formation during spinal fusion?"

✅ **Model Answer (10 Marks):**

The spinal cage balances **patient safety** and **bone formation** through a carefully structured biomimetic design:

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1. Patient Safety (Load Bearing + Structural Support)

- **Titanium Outer Shell:** Placed on the **apophseal ring** (strong cortical bone) to bear immediate post-op loads. Titanium is strong, stiff, and screw-fixable.
- **Slot-in Mg Core:** Resorbable insert aligned with softer **cancellous bone**, degrades over time to shift load to healing bone = **adaptive modulus**.
- **Chitosan Coating:** Controls magnesium degradation to avoid premature structural loss.
- **Personalised Core Options:** Surgeons can choose fast/med/slow degrading Mg depending on patient bone health.
- **Antimicrobial Coatings:** Optional silver/copper reduces infection risk without hindering bone regrowth (if dose-controlled).

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2. Design Features that Stimulate Bone Formation

- **Porous Structure:** Enhances nutrient/blood flow – critical when Mg degrades and flow is needed.
- **Concave Pores:** Mimic osteoclast pits → osteoblasts prefer curved zones, promoting **osteointegration**.
- **Submicron/Nanoscale Features:** Increase surface area for protein adsorption and cell attachment; proven to stimulate bone formation without chemicals.
- **Surface Coatings:**
  - **HA on Ti** promotes bone-on growth.
  - **Permanent surface charge** attracts growth factors.
- **rhBMP-2 Graft Chamber:** Central chamber holds BMP to actively stimulate new bone growth (**osteoinduction**).

Conclusion:
By combining **safe load-sharing mechanics**, **controlled degradation**, and **bioactive features**, the cage design ensures both structural safety and optimal bone formation during healing.

When should the change in modulus occur? - - - After initial bone ingrowth begins, typically within 6-12 weeks post-operation. Modulus should gradually shift as new bone forms, allowing progressive load transfer from cage to bone, aligning with natural bone healing stages. The change should be synchronised with osteogenesis, not before, to avoid loss of structural support.

"Explain how the design features of the spinal cage promote bone formation."

✅ **Model Answer (10 Marks):**

The spinal cage uses multiple **biomimetic and bioactive features** to enhance bone formation through osteoconduction, osteointegration, and osteoinduction:

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1. Porous Structure

- Provides channels for **nutrients, blood, and bone cells** to migrate and proliferate.
- Supports **vascularisation** and **waste removal**, both critical for healing.
- Especially important when Mg inner core degrades – maintains flow and avoids necrosis.

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2. Concave Pores (Biomimetic Cues)

- Mimic **osteoclast resorption pits** found in natural bone remodelling.
- **Osteoblasts prefer curved surfaces**, even without mechanical loading.
- Stimulate bone formation through geometry alone – shown effective in muscle pouch models.

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3. Submicron/Nanoscale Surface Features

- Created via **acid/alkaline etching** of the titanium outer shell.
- Increase **surface area**, improving **protein adsorption** and **cell attachment**.
- Shown to trigger bone growth even without added chemicals or growth factors.

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4. Surface Coatings

- **Hydroxyapatite (HA)**: Mimics natural bone mineral, enhances bone-on-growth on Ti or PEEK.
- **Nanoscale Ti on PEEK** (if used): Combines bioactivity and radiolucency.
- **Permanent surface charge** attracts endogenous BMPs and signalling proteins to the implant.

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5. Biological Additions

- **rhBMP-2** added to the **central graft chamber** → triggers osteoinduction.
- Stimulates osteoblast activity and formation of new bone tissue, even in poorly vascularised zones.

The spinal cage is designed to achieve **mechanical safety** while promoting **bone regeneration** through a biomimetic, layered structure.

- **Outer shell (Titanium):** 3D printed Ti interfaces with the **apophseal ring**, the load-bearing cortical region of the vertebra. It offers immediate strength and supports screw fixation.
- **Inner core (Magnesium alloy):** Positioned against the cancellous bone, the Mg insert gradually degrades to **transfer load to healing bone** — this is the **adaptive modulus** strategy, which follows **Wolff's Law**.
- **Chitosan coating**: Slows Mg degradation to match the bone healing timeline, maintaining support during early fusion.
- **Porosity and concave pores**: Scaffold-like geometry enables vascularisation and encourages osteoblast attachment. Curved surfaces mimic resorption pits, boosting osteointegration.
- **Nanoscale surface features**: Created by etching Ti, these promote protein adsorption and osteoinduction.
- **Central chamber with rhBMP-2**: Encourages bone formation even in poorly vascularised areas.
- **Antimicrobial coatings** (e.g. silver): Prevent infection while remaining biocompatible.

Monitoring bone fusion via imaging is essential in spinal surgery to confirm that new bone is forming within and around the cage. Without this, surgeons cannot assess whether the fusion is progressing or if intervention is needed.

However, **metal implants like titanium are radiopaque**, creating artefacts on X-ray and CT scans that obscure bone growth. Conversely, **materials like magnesium (Mg) and PEEK are radiolucent**, allowing better visibility.

The spinal cage uses a **hybrid imaging strategy**:

- **Titanium shell** provides mechanical support and anchors the cage but is radiopaque.
- **Magnesium core** is radiolucent and **gradually degrades**, which makes the central chamber more visible over time.

This creates an **adaptive radiolucency**: initially, the surgeon sees the implant location; later, as Mg degrades, the newly forming bone becomes visible on imaging.

This is crucial during the **6–12 week post-op period**, when clinicians assess whether fusion is occurring successfully or if the patient may need additional support.

Improved visibility means better decisions, earlier detection of non-fusion, and reduced unnecessary revision surgeries — directly improving patient safety and outcomes.

In the spinal cage design, the combination of **Titanium (Ti)** and **Magnesium (Mg)** offers the best balance of mechanical strength, biological performance, and radiographic monitoring — all of which contribute to safer and more effective spinal fusion.

Titanium (Ti) is ideal for the outer shell because:

- It is strong, stiff, and ideal for interfacing with cortical bone at the apophseal ring.
- It can be 3D printed with lattice porosity, and post-processed (acid etching) to create submicron roughness that enhances osteoblast adhesion and integration.

Magnesium (Mg) is used for the inner core because:

- It is **resorbable and radiolucent**, enabling **adaptive modulus** and visibility of bone formation on X-rays.
- It is bioactive, stimulating osteogenesis as it degrades.
- A **chitosan coating** slows degradation to match the rate of bone healing, ensuring safe load transfer.

Compared to alternatives:

- **PLA** degrades to acids, is not radiolucent, and has poor biocompatibility.
- **PEEK** is radiolucent but hydrophobic and difficult to integrate with bone.
- **CFR-PEEK** adds strength, but can't be 3D printed and may pose fibre migration risks if exposed.

Overall, the Ti-Mg combination enables:

- Early load-bearing strength
- Controlled degradation and load transfer
- Visibility of healing
- Better fusion outcomes and reduced need for revision surgery.

CASE STUDY 5 – KNEE

- Knee osteoarthritis (KOA) is a common type of arthritis that occurs when the cartilage in the knee joint wears down over time
- The global prevalence of KOA is 16% for people aged 15 and older, and 22.9% for people aged 40 and older. In the UK, an estimated 5.4 million people have KOA
- KOA is a long-term condition that can't be cured, but there are many ways to treat its symptoms. Pain from KOA can be severely debilitating and common treatments such as anti-inflammatory medication and steroid injections have limited efficacy

Summary of Case Study 5: GAE for KOA

The Clinical Problem:

- **Knee Osteoarthritis (KOA)** causes cartilage breakdown, inflammation, and pain.
- Inflammation leads to the growth of **new blood vessels and sensory nerves** inside the joint (**neoinnervation**).
- This makes movement painful as nerves now invade sensitive inflamed regions.

The Treatment – Genicular Artery Embolisation (GAE):

- **Minimally invasive** procedure done by interventional radiologists.
- Involves injecting **microspheres (embolic agents)** into the **genicular arteries** that feed these new vessels.
- The beads **block blood flow**, causing **ischaemia** → this **kills blood vessels and nerves**, stopping pain signals.

🧠 *"Kill the blood vessels → Kill the nerves → Stop the pain."*

What is the genicular artery embolization (GAE) and how it can be used to treat pain in knee osteoarthritis?

Genicular Artery Embolisation (GAE) is a **minimally invasive procedure** used to treat chronic knee pain, particularly in osteoarthritis (KOA), by **blocking abnormal blood vessels** in the knee joint.

- KOA causes **inflammation**, which promotes **new blood vessel and nerve growth** in the joint.
- These new nerves transmit pain signals.
- In GAE, **tiny microspheres (embolic agents)** are injected into **genicular arteries** to block these vessels.
- This cuts off blood supply to the new nerves, reducing pain — hence the phrase:

"Kill the vessels → kill the nerves → kill the pain."

LONGER ANSWER: Genicular Artery Embolisation (GAE) is a **minimally invasive endovascular procedure** used to relieve chronic pain associated with **Knee Osteoarthritis (KOA)**. It works by **intentionally blocking the small arteries (genicular arteries)** that supply blood to inflamed tissues within the knee joint. The process is performed by an **interventional radiologist** under image guidance using **fluoroscopy or digital subtraction angiography**.

In KOA, inflammation stimulates the growth of **new blood vessels and sensory nerves** into areas of joint damage. These newly formed nerves contribute to chronic pain by enhancing pain signalling within the joint. GAE targets these abnormal vessels using **particulate embolic agents**, which are injected into the genicular arteries. The beads block blood flow (ischaemia), which in turn **cuts off the nutrient supply to these nerves**, resulting in **nerve death and pain relief**.

This approach can significantly reduce pain and improve function in patients, especially those who are **not candidates for surgery** or wish to delay joint replacement. The treatment is **minimally invasive**, typically done under local anaesthesia with short recovery time.

Initial versions of GAE in Japan used a **mixture of antibiotics (e.g. imipenem-cilastatin)** that formed temporary embolic particles when mixed with contrast agents. These caused **transient ischaemia** and dissolved within hours, but safety concerns over off-label antibiotic use and allergic reactions limited their clinical acceptance.

As a result, newer **biomaterial-based embolic agents** (e.g. SakuraBead, based on alginate) are being developed to mimic this effect with **better safety, control, and regulatory approval**. These agents must have **predictable degradation**, **precise size**, and **low systemic toxicity**, while maintaining therapeutic effectiveness in blocking pathological vessels and nerves.

What are the key requirements for choosing the materials for the embolic agent suitable for GAE?

[10 marks]

Model Answer:

To be suitable for Genicular Artery Embolisation (GAE), an embolic agent must meet several **biological, mechanical, and functional criteria**. The key requirement is that it must **temporarily occlude blood flow** in the genicular arteries to reduce pain signalling by targeting inflamed nerves.

First and foremost, the material must be **biocompatible and non-toxic**, even during degradation. Since embolisation involves injecting the agent directly into blood vessels, **safety and predictability** are critical. The material must **not trigger immune responses**, inflammation, or clotting.

A **controlled degradation time** is essential — ideally around **1–3 hours**. This allows the agent to cause temporary ischaemia, sufficient to induce nerve damage but short enough to avoid long-term tissue damage or complications.

The agent must have **precise particle sizing**, typically between **200–300 µm**, to ensure targeted delivery without risking non-target embolisation. It must also be **mechanically robust**, with **compressibility** to pass through microcatheters and **elastic recovery** to expand and occlude the vessel.

Isobuoyancy is another key property: the beads should remain suspended in the contrast agent during injection, ensuring **uniform distribution** and avoiding settling.

Sterility and shelf stability are also required — the agent should be an **off-the-shelf product with a 3-year shelf life**, allowing widespread clinical use. Optional features like **radiopacity or dye** can improve visibility but should not interfere with flow or degradation.

Materials like **alginate hydrogels** meet many of these criteria and can be engineered with embedded enzymes for **predictable, controlled degradation**, making them highly suitable for mainstream GAE.

What is the best material for this application and why?

[10 marks]

Model Answer:

The best material currently identified for Genicular Artery Embolisation (GAE) is a **purified alginate hydrogel**, such as the one used in **SakuraBead®** microspheres. Alginate is a **naturally derived polysaccharide** extracted from seaweed, offering excellent **biocompatibility, tunability, and safety**.

Alginate is highly suitable for GAE because it forms **hydrogels via ionic crosslinking with calcium**, allowing precise control over particle size (200–300 µm), which is optimal for targeting the genicular arteries. The resulting beads are **soft and compressible**, able to pass through fine catheters without rupturing, and recover their shape upon injection to effectively occlude vessels.

What distinguishes alginate as a superior material is its ability to incorporate **alginate lyase enzyme** into the microsphere matrix. When rehydrated in saline just before use, the enzyme becomes active and begins to **degrade the microsphere in a controlled manner**. This allows a degradation time of 1–3 hours — ideal for causing temporary ischaemia and then allowing blood flow to return.

Additionally, alginate's degradation products are **non-toxic and safely excretable**, unlike other options such as antibiotic-based embolic agents, which have off-label usage risks. Alginate beads can also be sterilised using **gamma radiation**, are stable when dry, and can be stored as a **ready-to-use product**.

Overall, alginate offers a balance of **mechanical performance, controlled degradation, biocompatibility, and regulatory safety**, making it the best currently available material for mainstream clinical use in GAE.

Why does the size of the microsphere matter, and how are they made?

• Correct size ensures effective blocking of targeted arterioles (200–300 µm typical).
• Too small: risk of migration to non-target areas.
• Too large: may not reach or block the intended capillaries.

• Manufactured via crosslinking in calcium solution using methods such as:
  - Electrostatic droplet generation
  - Jet cutting
  - Microfluidics

• Enzyme (alginate lyase) embedded to control degradation rate post-delivery.

Q5: Explain the compressibility and isobuoyancy of the microspheres and why they are important.

• Compressibility: Allows microspheres to deform and pass through catheters easily without rupturing.
• Viscoelastic nature helps them regain shape and lodge in vessels.
• Important for safe delivery without fragmentation.

• Isobuoyancy: Microspheres match the density of contrast agent solution.
• Prevents settling or floating, ensuring even suspension and consistent flow during injection.

Q6: Why are some agents dyed or radiopaque? What are the pros and cons?

• Radiopaque microspheres help track embolisation under X-ray imaging.
• Dyed agents can confirm deployment visually.

Advantages:
• Improves targeting accuracy and monitoring.
• Confirms procedural success in real time.

Disadvantages:
• May introduce cytotoxicity or reduce biocompatibility.
• Regulatory approval may be harder due to added substances.

Q7: How would you make the embolic agent?

• Start with purified alginate solution.
• Mix with calcium ions to initiate crosslinking and form hydrogel microspheres.
• Use microfluidics or droplet generator for size control (200 ± 50 µm).
• Embed alginate lyase enzyme in the dry state (frozen in matrix).
• Dry and sterilise using gamma radiation.
• Package as dehydrated microspheres.
• Rehydrate in contrast agent at point of use to achieve isobuoyancy.

Compare dyed and radiopaque embolic microspheres in the context of Genicular Artery Embolisation (GAE). Discuss their advantages, disadvantages, and impact on clinical use.

[10 Marks]

✅ **Model Answer:**

In Genicular Artery Embolisation (GAE), microspheres are injected into the genicular arteries to block blood flow to pain-transmitting nerves. For safe and effective delivery, **visibility of the embolic agent** is crucial. This can be achieved by either **dyeing** the microspheres or making them **radiopaque**.

🔹 **Dyed Microspheres**

Advantages:

- Easy to see in the syringe or tubing.
- Confirms suspension and mixing quality before injection.
- Does not alter the density of the particles, maintaining **isobuoyancy**, which ensures even distribution during delivery.

Disadvantages:

- Not visible under X-ray or fluoroscopy.

- Only useful before or during initial injection — once injected, cannot be tracked in the body.
- Dye must be **biocompatible** and stable in saline/contrast agent.

🔹 **Radiopaque Microspheres**

Advantages:

- Visible under X-ray/fluoroscopy, especially **post-procedure**.
- Useful in oncology embolisation (e.g. tumour tracking) where permanent visibility is helpful.
- Can confirm site of embolisation *after* injection.

Disadvantages:

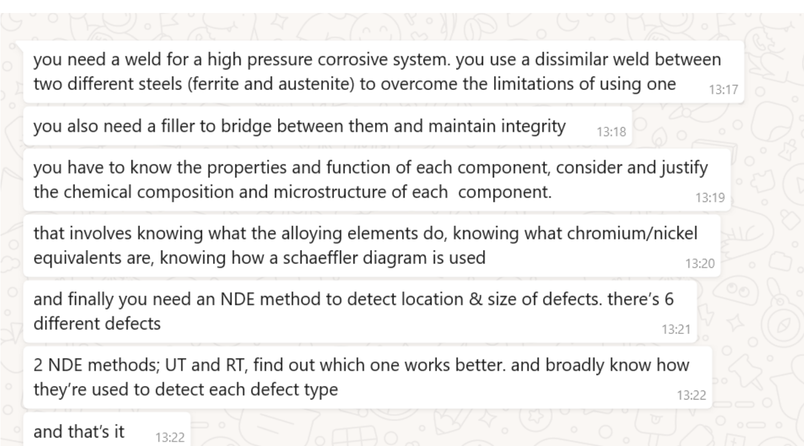
- Radiopaque materials (e.g. barium, tantalum) increase density → **cause sedimentation** in suspension.
- Poor suspension control can lead to **inconsistent dosing** or catheter blockages.
- In GAE, these are less helpful because embolisation is temporary, and tracking post-op is not critical.

✅ **Conclusion:**

For GAE, **dyed microspheres are preferred** because they maintain **uniform flow**, support dose control, and match the temporary, targeted nature of the treatment. Radiopaque agents, while useful in some interventional radiology contexts, present drawbacks in **buoyancy and precision** that make them less suitable for this minimally invasive pain relief procedure.

WELDING CASE STUDY

This case study focuses on a dissimilar transition weld between a ferritic low-carbon steel (SA508) and an austenitic stainless steel (Type 305), using a suitable weld filler (Type 309). The aim is to ensure mechanical compatibility and corrosion resistance at the weld zone while applying appropriate Non-Destructive Evaluation (NDE) methods to ensure volumetric weld integrity.



What are the material properties of a filler?

A good weld filler should:

- Match or exceed the strength and corrosion resistance of the weakest parent material.
- Be compatible in terms of chemical composition to avoid cracking or unwanted phase formation.
- Control the Ferrite Number (FN) to ensure an acceptable balance of austenitic/ferritic phases. For example, FN = 10 is ideal for crack and corrosion resistance.
- Type 309 is suitable: UTS ≈ 590 MPa, Elongation = 40%, Cr = 25%, Ni = 12%.

An ideal weld filler must provide structural continuity and resist cracking or corrosion at the dissimilar weld joint. Its key properties include:

- **Mechanical compatibility: It should match or exceed the strength and elongation of the weaker parent metal.**
- **Chemical compatibility: The composition should prevent brittle phase formation (e.g. sigma phase) and match thermal expansion.**
- **Phase balance: Controlled Ferrite Number (FN) is critical. FN ≈ 10 balances ductility and crack resistance.**
- **Example – Type 309 filler:**
    - **UTS ≈ 590 MPa**
    - **Elongation ≈ 40%**
    - **Cr ≈ 25%, Ni ≈ 12% This filler bridges ferritic and austenitic steels effectively and resists hot cracking.**

- **Q2: What is material selection process?**
- Choose ferritic carbon steel with: Yield > 550 MPa, UTS 650-950 MPa, Elongation >15% (e.g. SA508).
- Choose austenitic stainless steel with: Yield > 200 MPa, UTS > 550 MPa, Elongation >25% (e.g. Type 305).
- Use Schaeffler Diagram: Calculate Cr and Ni equivalents → Plot Points A (SA508), B (Type 305), D (filler = Type 309).
- Dilution of parent + filler gives Point E → ideal FN = 10-11.

The process involves selecting base metals and filler that are structurally and metallurgically compatible:

- **SA508 (ferritic low-carbon steel)**:
    - Yield > 550 MPa, UTS 650–950 MPa, Elongation >15%
- **Type 305 (austenitic stainless steel)**:
    - Yield > 200 MPa, UTS > 550 MPa, Elongation >25%
- **Filler – Type 309**:

    - Selected using **Schaeffler Diagram**

- Steps:
    1. Calculate Cr_eq and Ni_eq.
    2. Plot base metals and filler.
    3. Estimate dilution and plot resulting weld metal (Point E).
    4. Ensure FN ≈ 10 for a stable austenitic-ferritic mix. This ensures mechanical integrity and corrosion resistance at the weld.

3: What is the best welding process?

- Use TIG (Tungsten Inert Gas) welding:
    - Suitable for precise, narrow joint gaps and steep fusion angles.
    - Inert gas prevents oxidation.
    - Minimal dilution and precise control of heat input, essential for dissimilar metals.
- **High precision**: Ideal for narrow joint gaps and steep fusion angles.
- **Controlled heat input**: Prevents excessive dilution or phase transformation.
- **Inert gas shielding**: Prevents oxidation and contamination.
- **Minimal spatter and distortion**: Suitable for dissimilar metals with different expansion coefficients. TIG enables fine control needed when joining reactive and heterogeneous materials.

Tungsten Inert Gas (TIG) welding is preferred for dissimilar welds such as SA508 (ferritic steel) to Type 305 (austenitic stainless steel) due to its high precision and metallurgical control. Dissimilar metals present challenges like different melting points, thermal expansion rates, and risk of intermetallic phase formation.

Key advantages of TIG:

- **Precise heat input: TIG uses a non-consumable tungsten electrode, allowing focused energy input. This reduces dilution and preserves intended microstructure.**
- **Narrow, controlled weld pool: Ideal for small joint gaps and steep bevels typical in dissimilar transitions.**
- **Inert shielding gas (argon): Prevents oxidation and contamination, maintaining weld quality and reducing post-processing.**
- **Minimal spatter and distortion: Important since ferritic and austenitic steels expand differently — TIG avoids warping and stress accumulation.**

Why not use MIG or SMAW?

- **These have higher heat input, more spatter, and less precision — increasing risk of cracking or excessive dilution.**

Q4: What is the Non-Destructive Evaluation (NDE)?

- Inspection method to detect flaws without damaging the part.
- 7 types: Flaw detection, leak detection, metrology, microstructure evaluation, mechanical/physical property evaluation, stress/dynamic response, and location detection.
- Ensures the weld is free from critical defects and fit for service.

Non-Destructive Evaluation (NDE) is a group of inspection methods used to detect internal or surface flaws in welded joints without impairing the material. It ensures that welded components meet quality, strength, and safety requirements.

In this case, a dissimilar weld joins **SA508 (ferritic)** to **Type 305 (austenitic)** steel using **Type 309 filler**. These materials have different thermal and metallurgical behaviours, increasing the risk of **defects such as lack of fusion, cracks, or porosity**.

NDE ensures weld integrity by:

- Identifying internal flaws (e.g. porosity, centreline cracks) that compromise strength.
- Locating surface-breaking defects (e.g. toe cracks) that initiate fatigue failure.
- Ensuring the joint can withstand service stresses without premature failure.

NDE types used:

- **Ultrasonic Testing (UT):** Detects internal flaws using sound waves — ideal for detecting LORP, LOSWF, root cracks.
- **Radiographic Testing (RT):** Uses X-rays to identify voids and porosity.

Q5: What are the different types of NDE methods?

Key categories include:

- **Volumetric techniques**:
    - **Radiographic Testing (RT)**: Detects internal voids and porosity using X-rays.
    - **Ultrasonic Testing (UT)**: Uses high-frequency sound to detect cracks and fusion issues.
- **Surface techniques**:
    - Dye Penetrant, Magnetic Particle Testing
- **Advanced techniques**:
    - Acoustic Emission, Eddy Current, Thermography These methods are selected based on defect type, location, and material properties.

Q6: How to use the Schaeffler Diagram?

The Schaeffler Diagram predicts the weld metal's microstructure based on Cr_eq and Ni_eq:

- Formulas:

- $Cr_{eq} = Cr + Mo + 1.5Si + 0.5Nb$
- $Ni_{eq} = Ni + 30C + 0.5Mn$

- Steps:
  1. Compute Cr_eq and Ni_eq for SA508 (Point A), 305 (Point B), 309 (Point D).
  2. Predict unfilled weld zone (Point C).
  3. Estimate dilution of base and filler → Point E.
  4. Confirm Point E lands in region with FN ≈ 10. This ensures the weld resists hot cracking and retains toughness.

Q7: What type of faults/defects can occur in welding?

Common weld defects include:

- **LORP (Lack of Root Penetration)**: Incomplete fusion at root, weakens joint.
- **LOSWF (Lack of Side Wall Fusion)**: Fusion fails at edge, causes crack initiation.
- **Toe Crack**: Surface crack from thermal stress.
- **Root Crack**: Crack at joint base, hard to detect.
- **Centreline Crack**: Along centre axis, often from shrinkage.
- **Porosity**: Trapped gas bubbles; weakens joint and leaks. Such defects compromise performance and must be detected during inspection.

Q8: What is ultrasonic testing and how does it work?

Ultrasonic Testing (UT) is a volumetric NDE method that:

- Uses piezoelectric crystals to emit sound waves (1–10 MHz).
- Waves travel through material; reflect from interfaces or flaws.
- Reflected echoes are captured and displayed to indicate depth and type of defect.
- Can detect both surface and sub-surface defects with high sensitivity.
- UT is portable, accurate, and widely used for weld inspection.

- Ultrasonic Testing (UT) is a volumetric NDE method that:
  - Uses piezoelectric crystals to emit sound waves (1–10 MHz).
  - Waves propagate through material and reflect off interfaces or defects.
  - These reflections create echoes which are interpreted as peaks. Defects appear as additional peaks or changes in signal timing.
  - Uses both longitudinal and transverse (shear) waves:
    - Longitudinal waves: faster, deeper penetration, less sensitive to small defects.
    - Transverse waves: slower, more sensitive to smaller defects (e.g. cracks).
- Can detect both surface and sub-surface defects with high sensitivity.

- UT is portable, accurate, and widely used for weld inspection.

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what defect can be detected by which NDE method?

Defect Type	Best Detection Method
Toe Crack	UT: transverse or creep waves
LOSWF	UT: transverse waves at 60°
LORP	UT: transverse waves at 45°
Root Crack	UT: transverse waves at 45°
Porosity	RT preferred, UT possible
Centreline Crack	UT: dual transverse probes

Each method is chosen to match flaw orientation and expected location for maximum sensitivity.

Using the Schaeffler Diagram, explain how you would select a suitable filler material for a dissimilar weld between SA508 and Type 305 stainless steel. Include why controlling the Ferrite Number is important.

The Schaeffler Diagram is used to predict the microstructure of weld metal in dissimilar metal welding by comparing chromium and nickel equivalents. It helps ensure phase balance and crack resistance in the final weld.

- **Cr_eq = Cr + Mo + 1.5Si + 0.5Nb**
- **Ni_eq = Ni + 30C + 0.5Mn**

Steps to Select Filler:

1. Calculate Cr_eq and Ni_eq for:
   - SA508 (ferritic low-C steel) → Point A
   - Type 305 (austenitic SS) → Point B
   - Type 309 filler → Point D (Cr ≈ 25%, Ni ≈ 12%)
2. Point C = weld without filler.
3. Estimate dilution between parent metals and filler → Point E.
4. Ensure Point E falls in a region with **Ferrite Number (FN) ≈ 10**.

Why FN Matters:

- FN controls the balance between austenite and ferrite in the weld.
- Too much austenite → risk of **hot cracking**.
- Too much ferrite → reduces ductility.

"A low ferrite number suggests larger proportion of austenite and hence better corrosion resistance, while a high ferrite number infers higher cracking resistance. Ideal FN for dilution of weld filler in this case = 10, which gives good balance of crack resistance and corrosion properties."Type 309 is chosen as filler because it bridges ferritic and austenitic compositions, stabilises FN, and prevents unwanted phase formation like sigma phase.

How UT Works:

- UT uses a **piezoelectric transducer** to emit high-frequency sound waves (1–10 MHz) into the material.
- These waves travel through the component and reflect off any interfaces or flaws.
- Reflected signals are converted back into electrical pulses and displayed as an **A-scan**, allowing inspectors to measure **distance, size, and location** of defects.
- Both **longitudinal** and **transverse waves** can be used, with angles typically between 45°–70° depending on defect orientation.

🔹 **Why It's Suitable for Welds:**

- Welds may contain **volumetric defects** (e.g. porosity, slag inclusions) and **planar defects** (e.g. cracks, lack of fusion).
- UT is highly sensitive to **planar flaws** like:
  - **Lack of Root Penetration (LORP)** – detected at 45° angle.
  - **Lack of Side Wall Fusion (LOSWF)** – best seen at 60°.
  - **Root cracks** and **centreline cracks** – using twin probes or focused waves.
- Unlike Radiographic Testing (RT), UT can detect **subtle orientation-dependent cracks**, and works even in thick, non-uniform sections.

🔹 **Advantages:**

- Portable and safe (no radiation).
- Detects defects in real time.
- Effective in both **ferritic and austenitic regions**, though some grain noise may occur in stainless steels.
- Can be used during or after welding to assess structural integrity.

✅ **Conclusion:**

UT is ideal for ensuring the quality of dissimilar welds by detecting critical flaws without damaging the component. Its ability to locate and size defects like LOSWF and root cracks makes it essential for structural weld verification in high-risk applications like nuclear or pressure vessels.

CAMSHAFT CASE STUDY

What is a damper mass?

A damper mass (or pendulum damper) is a high-density mass used to reduce torsional vibrations in a rotating system, such as a camshaft. It does this by oscillating in the opposite direction to the rotation, generating a damping torque that absorbs vibrational energy. This reduces component fatigue, ensures smoother engine operation, and extends part lifetime.

How does it work?

- The damper mass is housed in a carousel and held by two washers.
- As the carousel rotates clockwise, the damper mass swings anti-clockwise due to inertia.
- The resulting damping torque opposes engine vibration.
- The torque generated depends on mass, radius, roller size, and number of dampers.
- The vibration order (e.g., 4th order) is tuned to match engine harmonic frequencie

How should you choose the materials for the damper mass and the carousel that it sits in?

SO IN 2010 season, some materials have been made illegal by FIA regulations. This includes the current material for the camshaft pendulum dampers. The material must now be below a density of 18,100 kg/m3. Packaging of the carousel is restricted to 18mm diametrically and you may fit as many pendulum dampers as you like per carousel.

Materials should obey the FIA regulations which states that: - a) Magnesium based alloys; - b) Metal Matrix Composites; - c) Intermetallic materials; - d) Alloys containing more than 5% by weight of Beryllium, Iridium or Rhenium.

What is a damper mass and how does it work?

A **damper mass** (also known as a pendulum damper) is a mechanical solution used in rotating systems like a camshaft to reduce **torsional vibrations**. These vibrations occur due to firing pulses and mechanical irregularities in the engine cycle. Left unchecked, they can lead to premature wear, noise, and mechanical failure.

The damper mass is mounted inside a **carousel** and held by two washers. As the camshaft rotates clockwise, the mass wants to stay still due to inertia and swings in the **opposite direction**. This oscillating action generates a **damping torque** that counteracts the vibrational energy.

- The system targets a **specific vibration order** (e.g. 4th order) corresponding to harmonics of engine speed.
- The **torque generated** is influenced by the mass (m), radial position (R), eccentricity (e), roller radius (r), and number of dampers.
- By tuning geometry and mass, the damper can be tailored to reduce specific frequency bands of vibration.

This leads to:

- Reduced fatigue of the camshaft and associated components.
- Quieter operation and reduced NVH (Noise, Vibration, Harshness).

- Better reliability under F1 engine conditions.

✅ **How should you choose materials for the damper mass and carousel?**

🔹 **Damper Mass Requirements:**

- **High Density:** To generate significant torque within compact volume. Ideal density ≈ 17,000–18,500 kg/m³.
- **Hardness:** Needs to be hard enough to resist fretting and wear, but not so hard that it damages the carousel. Target ≈ 300–400 HV.
- **Yield Strength:** High enough to prevent deformation at high rotational speeds. Aim for >500 MPa.
- **Thermal and Oil Compatibility:** Must operate in hot, oil-rich environments without corrosion or degradation.

🔹 **Carousel (Housing) Requirements:**

- **Material Match:** Typically EN9 steel (tough, cost-effective, and heat treatable).
- **Hardness Compatibility:** Avoid mismatch that causes wear – ideal between 150–350 HV.
- **Fatigue Strength:** The carousel holds multiple dampers, so it must resist cyclic loading.
- **Machinability:** Tight tolerances and bore alignment are critical.
- **Operating Environment:** Must tolerate vibration, oil, and temperatures up to 140°C.

In short, material choice must balance density, strength, machinability, and surface compatibility.

✅ **What materials should you use and why?**

🔹 **Damper Mass – Tungsten Heavy Alloy (e.g. W-Ni-Fe or W-Ni-Cu)**

- **Density:** ~17,000–18,500 kg/m³ – higher than steel or nickel alloys, allows compact high-torque design.
- **Hardness:** 300–400 HV, which is compatible with EN9 carousel and avoids excessive wear.
- **Strength:** Yield strength > 700 MPa → handles high-speed inertial loads without plastically deforming.
- **Machinability:** Unlike Tungsten Carbide, heavy alloys can be CNC machined precisely.
- **Oil & Temp Resistant:** Chemically stable in oil and elevated temps found near the valvetrain.

🔹 **Why not other materials?**

- **Tungsten Carbide:** Too brittle and hard (>1,500 HV), risks damaging the housing.
- **Nickel Alloys (e.g. Inconel):** Lower density (~8,000–9,000 kg/m³), so much larger size needed.
- **Steel:** Also low in density, would require doubling volume or number of dampers.

Thus, **tungsten alloys offer the best balance** of density, machinability, hardness, and mechanical performance.

✅ **How might you make an individual damper mass?**

1. **Raw Material:** Start with W-Ni-Fe alloy rod stock (cut to near-final size).
2. **CNC Machining:** Use precision lathes to machine the mass to ~15 mm length, exact radius (R) as per torque formula.
3. **Fine Surface Prep:** Use grinding, polishing to remove burrs and improve finish. Optional: diamond polishing.
4. **Surface Coating (Optional):** Apply DLC, chrome, or PVD to reduce wear if needed.
5. **Quality Control:**
    - Mass tolerance: to within ±0.1g.
    - Dimensional inspection: ensure bore fits.
    - Hardness test: HV test to confirm compatibility with housing.

Carousel:

- Machined from EN9 steel.
- Features precision bores and slots for each damper.
- May be heat treated (tempered or normalized).
- Final step: dynamic balancing and lubricant application.

EXTRA: If the damper includes flat features or non-cylindrical geometry, **CNC milling** may also be used. This allows more flexible shaping compared to lathes alone. The part must be able to handle **high inertial loads and surface wear**, so coatings like DLC or chrome are applied depending on the application — e.g., **DLC for high-speed, oil-lubricated systems**.

✅ **How should we surface treat the damper mass to improve lifetime?**

The damper mass undergoes high-speed oscillation within a steel housing → surface treatment is critical to resist wear and fretting:

- **DLC (Diamond-Like Carbon):** Excellent wear and friction reduction. Biocompatible with EN9.
- **PVD/CVD Coatings:** Thin, hard ceramic-like coatings increase surface hardness.
- **Chrome Plating:** Adds corrosion resistance and wear resistance.
- **Heat Treatment:** Not typical for tungsten alloys but may be used on washer/housing components.

- **Lubrication:** In-situ oil splash or added moly-based grease reduces metal contact.
- **Precision Finish:** A smoother finish ensures minimal surface abrasion over lifetime.

Combining coatings with good clearances and lubrication leads to longer-lasting, smoother damping performance.

1. **Raw Material:** Start with W-Ni-Fe alloy rod stock (cut to near-final size).
2. **CNC Machining:** Use precision **lathe** to machine the damper mass into a **cylindrical shape**:
    - Length = 15 mm
    - Radius optimised via torque calculation
    - Maintain tight tolerance (±0.1 mm)
    - **Alternatively**, milling may be used if damper has flats, slots or non-cylindrical features.
3. **Fine Surface Prep:** Grinding and polishing to remove burrs and reduce friction. Optional: diamond polishing.
4. **Surface Coating (Optional):** Apply DLC, chrome, or PVD to reduce wear and improve lifespan.
5. **Quality Control:**
    - Mass check: within ±0.1g tolerance
    - Dimensional inspection (e.g., bore fitting)
    - Hardness test: to confirm compatibility with housing

Carousel:

- Material: EN9 steel, heat-treatable and tough.
- Machined to include **precision bores** and **slots** for washers.
- CNC milled or turned; final balancing and lubricant applied.
- May be surface-hardened and coated.

Why tungsten:

Tungsten heavy alloys like W-Ni-Fe are ideal for damper masses due to their extremely **high density (~18,000 kg/m³)**, allowing significant torque generation in a small form factor — critical for space-constrained F1 engine components. They offer a **hardness of 300–400 HV**, which is well-matched to the EN9 steel carousel and avoids wear from hardness mismatch.

The alloy also has a **high yield strength (>700 MPa)**, ensuring resistance to deformation under high-speed oscillations. Unlike brittle materials like **tungsten carbide**, tungsten heavy alloys are **machinable** via CNC turning/milling. This enables precise control over geometry and mass, critical for vibration tuning.

Alternatives like **Inconel** and **steel** are rejected due to **lower density**, which would require larger or multiple masses. **Tungsten carbide** is rejected for being **too hard and brittle**, risking housing damage.

Therefore, tungsten heavy alloy offers the **best balance of density, strength, and compatibility** for performance and manufacturability in this dynamic application.

🔹 **Damping Torque Equation:**

$$\text{Torque} = m \cdot \omega \cdot R \cdot 0.5 \cdot \frac{(R+e-25/r)}{1+(25/r)^2} \cdot n_{dampers}$$

Where:

- **m** = mass of damper
- **ω** = engine angular velocity (rad/s)
- **R** = distance from center to damper (mm)
- **e** = eccentricity
- **r** = roller radius
- **n_dampers** = number of dampers

Design Trade-offs:

- Higher **m**, **R**, or **number of dampers** → more torque
- Larger **r** (roller) → less torque

🔹 **Vibration Tuning Equation (for 4th order):**

$$\text{Order} = \left(\frac{R}{e(1+37.5/r^2)}\right)^{1/2}$$

- Choose **R, e, r** to target a desired harmonic (e.g. 4th order)
- Typically tuned to match camshaft harmonic torsional vibration

These equations allow iterative design to optimise torque and match target vibration order.

✅ How do the vibration and torque calculations work?

Damper Torque Equation: $\text{Torque} = m \cdot \omega \cdot R \cdot 0.5 \cdot \frac{(R+e-25/r)}{1+(25/r)^2} \cdot n_{dampers}$

- More mass or bores → Higher torque.
- Larger roller radius → Lowers torque.

Vibration Order Equation: $\text{Order} = \left(\frac{R}{e(1+37.5/r^2)}\right)^{1/2}$

- Target: 4th order vibration mode.
- Adjust R, e, and r to tune the system to engine harmonics.

How do you design to counter fretting / abrasion / wear in practice?

- **Material Selection:** Use wear-resistant materials.
- **Surface Engineering:** DLC, chrome, PVD/CVD coatings.
- **Lubrication:** Reduces friction and surface wear.
- **Friction Control:** Ensure clearance, good surface finish, and compatible hardness.

How do you design to counter fretting / abrasion / wear in practice?

To reduce damage at the damper-to-carousel interface, design includes:

- **Material Pairing:** Tungsten alloy mass (300–400 HV) with EN9 steel housing (200–350 HV) prevents harsh abrasive mismatch.
- **Surface Engineering:** Apply DLC or PVD coatings to reduce contact stress and micromovement-induced damage.
- **Oil Lubrication:** Ensures metal-to-metal contact is avoided.
- **Precision Machining:** Minimise clearances to control damper movement while avoiding binding.
- **Edge Rounding:** Avoid sharp transitions that promote fretting.
- **Dynamic Balancing:** Prevents localised overloading during engine operation.

By combining coatings, tight tolerances, compatible materials, and lubrication, fretting and wear can be effectively minimised.

Design goal: To design a tyre that reduces TRWP while keeping performance requirements, for a passanger car type.

Whats is TRWP? TRWP(tyre road wear particles) are tiny debris released into the environment and road, due to friction between the tyre and road.

What causes this friction? Smear and abrasion wear causes these TRWP. Smear wear happens when the temperature between road the tyre rises, causing the tyre compound to 'melt' and get stuck onto the road. While abrasive wear causes parts of the tyre compound to release as little particles.

What type of wear is better overall to reduce TRWP? Smear is better, as it is less damaging to the tyre. Therefore less damaging to the environment.

What is the solution? To descres TRWP we need good material, good tyre design and good manufacturing process and most importantly good end of life cycle. Materials: The tyre is made up of rubber + additives. Rubber part being FSBR (fictionalised styrene butane rubber) and Natural rubber. Fillers being: Silica and carbon black. Others: Soya oil for plasticer and 6PDD as antioxidant. Each serve part, to improve tyre's rolling resistance, reduced abrasion, and optimise longevity. All of which help reduce TRWP. Better performance+lifespan= less tyre replacement therefore less TRWP released.

Manufacturing process includes: assembly of tyre parts, cross-linking and vulcanisation.

Problems; 6PPD releases 6PDDQ which is toxic for marine life. Research on replacement for this has to still be done. The design of tyres are important as it contributes to the overall polluted environment today, and changing the materials or design can give a good outcome in the future.

End of life: most tyres end up in landfill. Some recycled. Therefore providing stricter rules on old car tyres is needed going forward, as well as material substitution. Note: Driving behaviour and road conditions greatly affect TRWP production therefore keep mindful.

Point of the case study: was to research how we could find better design or material selection to reduce TRWP released into the environment. As TRWP produce sludge, affective marine life as well as us. The supply demand is increasing, meaning greater problem at disposing of tyres.

1. What types of abrasion take place in practice?

There are two main types of abrasion in tyre wear: **smear wear** and **abrasive wear**.

- **Smear wear** occurs under low severity conditions (e.g., normal road driving), where the rubber compound softens and smears onto the surface. It is driven by **tribo-chemical degradation**, where frictional heat oxidizes the surface and forms a sticky protective layer. This reduces further crack propagation.
- **Abrasive wear** occurs during high-performance or aggressive driving, where **thermo-mechanical degradation** dominates. Cracks form at macro scales, and the tyre is worn away like sandpaper by road asperities, generating sharp and harmful **Tyre Road Wear Particles (TRWP)**.

Smear wear is the **preferred mode** as it generates larger and softer particles that tend to stay near the road, reducing airborne pollution and material loss.

✅ 2. What is the difference between smear wear and abrasive wear in tyres?

- **Smear wear** is a low-severity wear mechanism, involving softening and slight flowing of rubber due to heat and chemical changes. Micro-cracks form but do not propagate deeply. This leads to fewer particles released and less mass loss.
- **Abrasive wear**, in contrast, is high-severity and mechanical. The road acts as an abrasive tool, creating deeper cracks, sharp tears, and small TRWPs with high PM2.5 risk.

Smear wear is more desirable: it protects the rubber, prolongs tyre life, and reduces environmental TRWP impact. Therefore, modern tyre design focuses on materials and fillers that promote smear rather than abrasion.

✅ 3. Which wear process is best for tyre life?

Smear wear is best for tyre life and environmental performance.

- It occurs via soft plastic or viscoelastic deformation, which doesn't lead to material loss as quickly.
- It allows a protective layer to form on the rubber surface.
- Results in less tread loss and fewer TRWPs.

By contrast, **abrasive wear** leads to fast material loss and poor longevity. Promoting smear wear through **rubber elasticity**, **nanostructure design**, and **proper filler interaction** is key to optimising tyre durability.

✅ 4. How are these behaviours measured using lab tests?

Several **standardised lab tests** simulate real driving to evaluate tyre wear modes:

- **DIN Abrasion Test**: Measures mass loss by rubbing rubber against abrasive paper under load.
- **Rolling Resistance Test**: Evaluates energy loss during rotation.
- **Chips & Cuts Resistance**: Simulates sharp object impact to gauge abrasion.
- **TRWP Particle Analysis**: Microscopy and chemical profiling (e.g., FTIR, SEM, AFM) used to analyse wear debris size, shape, and composition.
- **Modulus mapping** (AFM): Helps map phase distribution and filler localisation in nanostructures, giving insight into wear mechanics.
- **Time-temperature superposition**: Predicts material behaviour under various speeds and temperatures.

These help identify whether a compound favours smear (good) or abrasive (bad) wear.

✅ 5. How does the impact of this relate to rolling resistance and wet grip?

Tyre performance is a **balance** between wear resistance, rolling resistance (RR), and wet grip (the "magic triangle").

- **Rolling Resistance** (fuel economy): Reduced by using unfilled NR or silica-filled SSBR blends with **low hysteresis** (energy loss) at low frequency.
- **Wet Grip** (safety): Improved by using filled, modified SBR with **high hysteresis** at high frequencies (cold/wet braking).
- Smear wear reduces frictional loss and favours **low RR**, but may slightly reduce grip.
- Abrasive wear causes fast tread loss, poor RR, and inconsistent wet grip.

Thus, **bi-continuous nanostructures** (e.g., filled SSBR + unfilled NR) are engineered to deliver **high toughness and optimal hysteresis** across driving scenarios.

✅ 6. What should the ideal compound look like and what does each ingredient add?

Ideal formulation balances toughness, elasticity, and environmental safety:

- **Polymers**:
    - **NR (70 phr)**: Elastic, low RR, encourages smear wear.

  - **Functional SSBR (30 phr)**: High wet grip, good filler dispersion.
- **Fillers**:
  - **Silica (50 phr)**: Reduces RR, boosts wet grip.
  - **Carbon Black (5 phr)**: UV protection, extra strength.
- **Coupling Agent (TESPT, 5 phr)**: Enhances filler-polymer bond.
- **Plasticiser (TDAE Oil, 20 phr)**: Improves processing, eco-safe.
- **Antioxidant (Alternative to 6PPD)**: Protects against cracking, less toxic.
- **ZnO Alternatives**: Reduce metal leaching in environment.
- **Accelerators & Sulfur**: Enable cross-linking during vulcanisation.

This recipe creates a **bi-continuous nanostructure** optimised for high wear resistance, low RR, and strong wet grip—while being safer for ecosystems.

✅ **7. If you use a blend of compounds, how should the materials appear in their microstructure and why?**

The best tyre tread compound exhibits a **bi-continuous microstructure**, where:

- **NR phase** remains unfilled for low hysteresis and flexibility.
- **SSBR phase** is highly filled with silica and carbon black to ensure toughness and wet grip.

This structure avoids phase separation and ensures:

- Good **filler localisation** (only in SSBR) via chain-end modification.
- Optimised viscoelastic response at both high and low frequencies.
- Reduced TRWP through promotion of smear wear and better energy dissipation.

Uniform low-temperature **vulcanisation** ensures consistent crosslinking and phase connectivity.

✅ **8. How are tyres made?**

Tyre Manufacturing Process:

1. **Mixing (Banbury Mixer)**: Rubber polymers + fillers + oils blended into a uniform compound.
2. **Sheeting & Cooling**: Compound flattened into sheets and cooled.
3. **Extrusion**: Rubber is shaped (tread profile) using dies.
4. **Tyre Building**:
   - Carcass built on a rotating drum.

- Beads, sidewalls, and tread added in layers.
- Steel belts added for reinforcement.

5. **Vulcanisation**:
   - "Green tyre" is placed in mould, heated under pressure.
   - Crosslinking via sulfur improves elasticity and strength.
6. **Finishing**:
   - Trimming, cooling, and quality control checks (X-ray, balancing).

This multistep process ensures a **durable, high-performance** tyre optimised for both road safety and sustainability.

6PPD – toxic but that's the only one that's currently

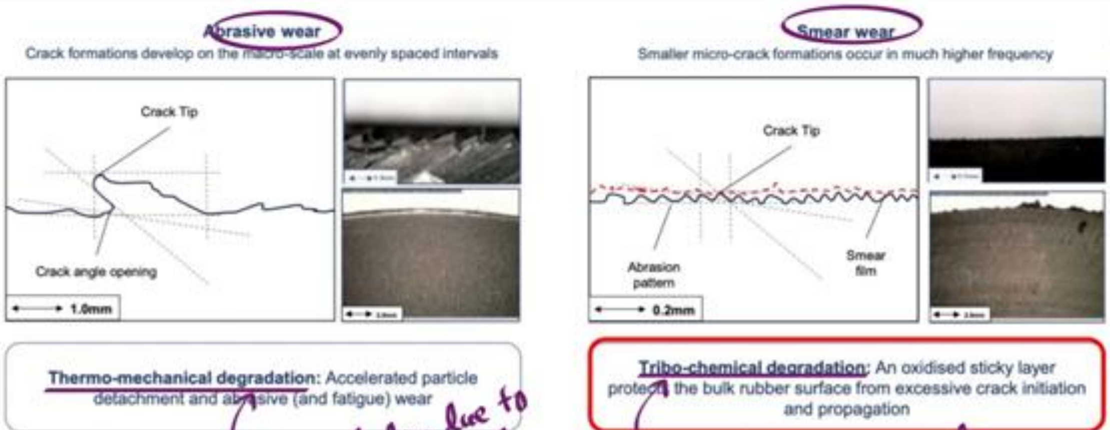
Difference between smear & abrasive wear

1. Abrasive vs Smear Wear

Smear Wear	Abrasive Wear
▪ Generates Sticky rubber debris. (Debris stay on the surface rather than instantly removing them into the environment)**[19]** ▪ Wear particles have high oxygen content **[19]**, which can cause degradation due to oxygen exposure. ▪ De-crosslinks the rubber making it weak. ▪ Even though, smear wear weakens the rubber initially, oxidation and filler interactions, can helps to stabilise the material, over time. **[19]** ▪ Dominates at higher temperature.	▪ Most common type of rubber wear. ▪ Wear happens due to contact between a hard surface and a soft surface. (Road and tyre surface) ▪ Caused by micro-cutting and fracture. ▪ So, more wear loss. ▪ Mainly depends on surface (road surface and tyre surface) roughness than, other aspects like energy loss. ▪ Doesn't' alter the molecular structure of rubber.

Table 9: Smear wear vs Abrasive wear [19]

Overall, smear wear is preferable for tyres because it produces sticky wear particles, which are not instantly lost to environment. Although smear wear weakens the rubber, filler materials can help reinforce it. While abrasive wear, caused by cutting and stretching, which leads to material loss that cannot be reversed.



(degradation due to mechanical stress & high T

(degradation due to chemical reaction + mechanical action